



Alberta Public Health Disease Management Guidelines

Powassan Encephalitis



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Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Isolation of Powassan encephalitis (POW) virus from an appropriate clinical specimen (e.g., serum, cerebrospinal fluid [CSF])^(B)

OR

- Detection of POW virus antigen in tissue, blood or CSF

OR

- Detection of POW viral nucleic acid (e.g., PCR) in an appropriate clinical specimen (e.g., plasma, CSF, serum)^(B)

OR

- Seroconversion or significant change between acute and convalescent phase POW Hemagglutination Inhibition Assay (HAI) antibody titres ideally taken at least two weeks apart from an appropriate clinical specimen (e.g., serum)^(B) and confirmed by Plaque Reduction Neutralization Assay (PRNT)^(C)

Probable Case

Clinical illness^(A) and one of the following:

- Seroconversion or significant change between acute and convalescent phase POW HAI antibody titres ideally taken at least two weeks apart but not confirmed by PRNT^(C)

OR

- Stable elevated serial HAI titres^(D) to POW that occur during a period when and where arboviral transmission is likely^(E)

OR

- Single elevated HAI titre^(D) to POW that occurs during a period when and where arboviral transmission is likely^(E)

^(A) Clinical illness is characterized by a febrile illness of variable severity associated with neurological symptoms ranging from headache to aseptic meningitis or encephalitis. Arboviral encephalitis cannot be distinguished clinically from other central nervous system (CNS) infections. Symptoms can include headache, confusion or other alteration in sensorium, nausea and vomiting. Signs may include fever, meningismus, cranial nerve palsies, paresis or paralysis, sensory deficits, altered reflexes, convulsions, abnormal movements and coma of varying degree.

^(B) Refer to the [National Microbiology Laboratory \(NML\) Guide to Services](#) for current specimen collection and submission information.

^(C) Seroconversion indicates recent infection with a flavivirus (e.g., dengue fever, California serogroup viruses or West Nile virus) but cannot pinpoint which one due to antibody cross-reactivity. These are considered probable cases. PRNT confirmatory testing may be requested at the NML through the Public Health Laboratory (ProvLab).

^(D) A single elevated antibody titre of $\geq 1:320$ by HAI suggests recent infection. It is recommended that a second specimen be collected. A second laboratory result with a stable (unchanged/static or $\leq$ twofold rise) elevated antibody titre is suggestive of recent infection.

^(E) POW exposure is most likely to occur from June to September in Canada and the United States and from spring to early summer in Russia. ⁽¹⁾

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Reference 2 applies to this section.

POW virus is a member of the *Flavivirus* family. POW cannot survive outside of the host and is sensitive to drying as well as moist and dry heat.

Clinical Presentation

Reference 2 applies to this section.

POW is a rare cause of acute viral encephalitis. Infection ranges from asymptomatic to mild to severe illness. Symptoms, when present, usually begin suddenly seven to 14 days following infection and include headache, fever, nausea and vomiting, stiff neck and sleepiness. As the disease progresses, more severe symptoms develop, such as breathing distress, tremors, confusion, seizures, coma, spastic paralysis and sometimes death. Individuals who recover may have residual neurological problems. The case fatality rate is 0.3–60%.

Diagnosis

Diagnosis of POW is made through rapid serologic assays such as IgM capture ELISA and IgG ELISA. In early infection IgM antibody is more specific, while later in infection, IgG antibody is more reactive. Viral isolation and identification have been useful in defining viral agents in serum, CSF, and mosquito vectors.

Treatment

- Antibiotics are not effective.
- No effective antiviral treatment has been identified.

Reservoir

The reservoir is most often blood feeding ticks. Humans and domestic animals can develop clinical illness but are usually dead-end hosts as they do not produce significant viremia and do not contribute to the transmission cycle.⁽³⁾

The host range for POW includes humans, woodchucks, coyotes, foxes, raccoons, skunks, and domestic cats and dogs.⁽⁴⁾ The reservoir is ticks or combination of tick and mammal.

Transmission

Infection is transmitted to humans through the bite of an infected tick, most commonly *Ixodes marxi*, *I. cookie*, and *Dermacentor andersoni*.⁽⁴⁾

Incubation Period

The incubation period of POW is usually seven to 14 days.

Period of Communicability

POW is also not transmitted directly from person-to-person. A tick infected at any stage remains infected for life. Viremia in a variety of vertebrates may last for several days. In humans, viremia typically lasts seven to 10 days.

Incidence

Reference 2 applies to this section.

General

The first recognized cases of POW in the United States (US) occurred in New Jersey in 1970. From 1958 through 1998, a total of 27 cases of human POW were reported from Canada and the north-eastern US (Massachusetts, New Jersey, New York). From 1999 to 2001, four cases were reported in Maine and Vermont.

Refer to the [U.S. Centers for Disease Control website](#) for more information.

Canada

POW virus was first isolated in 1958 from a child who died in Powassan, Ontario.⁽²⁾ POW is not a nationally notifiable disease, and the annual incidence is not known.

Refer to [Public Health Agency of Canada website](#) for more information.

Alberta

There were no cases of POW reported in Alberta between 1994 and 2020.

Public Health Management

Key Investigation

- Identify the potential source of the infection based on:
 - history of travel to endemic areas,
 - mode of transmission, and
 - incubation period.
- Assess for history of tick bites.

Management of a Case

- Supportive care, including IV fluids and ventilation.
- Symptomatic treatment.

Management of Contacts

The infection is not passed from person-to-person.

Preventive Measures

- Avoid tick-infested areas when possible (see [Incidence](#)).
- Use personal protective measures to minimize exposure and reduce the risk of tick bites, as follows.
 - Wear long sleeved shirts and long pants that are tight at the wrist and ankles or tucked into gloves or socks.
 - Light coloured clothing can aid in the detection of ticks that have not yet attached.
 - Wear a hat where contact with vegetation cannot be avoided, such as in dense woods, high grasses, or thickets.
 - Apply permethrin to tents, pant legs, and sleeves. Do not apply directly to skin.
 - Apply insect repellent containing 20–30% N,N-diethyl-3-methylbenzamide (DEET) (adults only) or up to 20% Picardin (anyone over six months of age) to the skin.
 - DEET is not recommended for use on children under six months of age.
 - For children age two to 12 years, insect repellent with < 10% DEET may be used up to three times per day.
 - For children six months to two years of age, insect repellent with < 10% DEET may be used once per day.
- If working or playing in a tick-infested area, daily inspection and prompt removal can prevent transmission of disease.
- Laboratory safety – Arboviruses require biosafety level 3 practices.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	• Updated Template • Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology • Key Investigation section moved to Public Health Management (formerly called Control) • Updated web links
	Incidence	• Updated Canadian and Alberta data and added hyperlinks to CDC and PHAC websites
	Preventive Measures	• Re-worded to align with other tick-borne disease guidelines.

References

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- (3) CDC. Guidelines for arbovirus surveillance programs in the United States. Department of Health and Human Services. Colorado: April 1993.
- (4) Public Health Agency of Canada. Infectious substances: Powassan encephalitis virus. Office of Laboratory Security. Material Safety Data Sheet. March 2001.
- (5) Public Health Agency of Canada (PHAC). Statement on Personal Protective Measures to Prevent Arthropod Bites - Update. Can Commun Dis Rep 2012;38(ACS-3).